

# Solutions to Problem Set 3

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## Problem 1

1. The Bellman equation is given by

$$V(B) = \max_C \left\{ \frac{C^{1-\sigma}}{1-\sigma} + \beta V(RB - C) \right\}.$$

2. Substitute the conjecture into the RHS of the equation and take the FOC:

$$C^{-\sigma} = \beta\gamma(RB - C)^{-\sigma}.$$

Simplify to express the optimal consumption function as

$$C = \frac{1}{1 + (\beta\gamma)^{\frac{1}{\sigma}}} RB.$$

Substitute this result and the value function into the Bellman equation:

$$\alpha + \gamma \frac{B^{1-\sigma}}{1-\sigma} = \frac{1}{1-\sigma} \left[ \frac{1}{1 + (\beta\gamma)^{\frac{1}{\sigma}}} RB \right]^{1-\sigma} + \beta\alpha + \beta\gamma \left( \frac{(\beta\gamma)^{\frac{1}{\sigma}}}{1 + (\beta\gamma)^{\frac{1}{\sigma}}} \right)^{1-\sigma} \frac{(RB)^{1-\sigma}}{1-\sigma}.$$

The fact that there are only constant terms and  $B^{1-\sigma}$  confirms our initial conjecture. It also follows that  $\alpha = 0$ . Combining other terms, we get

$$\gamma = [1 + (\beta\gamma)^{\frac{1}{\sigma}}]^\sigma R^{1-\sigma}.$$

Take both sides of the equation to the power  $1/\sigma$  and solve for  $\gamma$ :

$$\gamma = \frac{R^{1-\sigma}}{\left[1 - \beta^{\frac{1}{\sigma}} R^{\frac{1}{\sigma}-1}\right]^{\sigma}}.$$

3. The HJB equation is given by

$$\rho V(B) = \max_C \left\{ \frac{C^{1-\sigma}}{1-\sigma} + V'(B)(rB - C) \right\}.$$

4. Conjecture again that  $V(B) = \alpha + \gamma \frac{B^{1-\sigma}}{1-\sigma}$ . Substitute this function into the RHS of the HJB and take the FOC:

$$C^{-\sigma} = \gamma B^{-\sigma} \quad \Rightarrow \quad C = \gamma^{-\frac{1}{\sigma}} B.$$

Substitute this solution and the value function into the HJB:

$$\rho\alpha + \rho\gamma \frac{B^{1-\sigma}}{1-\sigma} = \frac{\gamma^{1-\frac{1}{\sigma}} B^{1-\sigma}}{1-\sigma} + \gamma B^{-\sigma} \left( rB - \gamma^{-\frac{1}{\sigma}} B \right).$$

Collect terms with  $B^{1-\sigma}$  and simplify:

$$\gamma = \left[ \frac{\sigma}{\rho - (1-\sigma)r} \right]^{\sigma}.$$

Finally, notice that  $\alpha = 0$ .

## Problem 2

1. The price  $p_{t-1}$  is taken as given in period  $t$  and summarizes all relevant information for firm's optimal choice in that period. Therefore, it is a state variable.
2. In period  $T$ , the firm solves

$$\max_{p_T} \left\{ -p_T^2 - \gamma(p_T - p_{T-1})^2 \right\}.$$

The first-order condition implies  $p_T = \frac{\gamma}{1+\gamma}p_{T-1}$  and

$$V_T(p_{T-1}) = -p_T^2 - \gamma(p_T - p_{T-1})^2 = -\frac{\gamma}{1+\gamma}p_{T-1}^2.$$

3. In period  $T - 1$ , the firm's problem is

$$\max_{p_{T-1}} \left\{ -p_{T-1}^2 - \gamma(p_{T-1} - p_{T-2})^2 + \beta V_T(p_{T-1}) \right\}.$$

Take the first-order condition

$$-p_{T-1} - \gamma(p_{T-1} - p_{T-2}) - \beta \frac{\gamma}{1+\gamma} p_{T-1} = 0$$

and obtain

$$p_{T-1} = \frac{\gamma}{1+\gamma+\frac{\beta\gamma}{1+\gamma}} p_{T-2}.$$

It follows that

$$V_{T-1}(p_{T-2}) = -p_{T-1}^2 - \gamma(p_{T-1} - p_{T-2})^2 - \frac{\beta\gamma}{1+\gamma} p_{T-1}^2 = -\frac{\gamma(1+\frac{\beta\gamma}{1+\gamma})}{1+\gamma+\frac{\beta\gamma}{1+\gamma}} p_{T-2}^2.$$

4. Conjecture that  $V_t(p_{t-1}) = -a_t p_{t-1}^2$ . Then the period-t optimization problem is

$$\max_{p_t} \left\{ -p_t^2 - \gamma(p_t - p_{t-1})^2 - \beta a_{t+1} p_t^2 \right\}.$$

Take the first-order condition:

$$p_t + \gamma(p_t - p_{t-1}) + \beta a_{t+1} p_t = 0.$$

It follows that

$$p_t = \frac{\gamma}{1+\gamma+\beta a_{t+1}} p_{t-1}$$

and the value function is

$$V_t(p_{t-1}) = -p_t^2 - \gamma(p_t - p_{t-1})^2 - \beta a_{t+1} p_t^2 = -\frac{\gamma(1+\beta a_{t+1})}{1+\gamma+\beta a_{t+1}} p_{t-1}^2.$$

Thus, parameter  $a_t$  is determined recursively according to

$$a_t = \frac{\gamma(1 + \beta a_{t+1})}{1 + \gamma + \beta a_{t+1}}$$

with the initial condition  $a_{T+1} = 0$ .

5. To find the limiting value  $a$ , impose  $a_t = a_{t+1}$ , obtain

$$\beta a^2 + a(1 + \gamma - \gamma\beta) - \gamma = 0$$

and choose a positive root of this equation

$$a = \frac{1}{2\beta} \left( \sqrt{(1 + \gamma - \gamma\beta)^2 + 4\beta\gamma} - (1 + \gamma - \gamma\beta) \right).$$

The Bellman equation for an infinite period problem is

$$V(p) = \max_{p'} \left\{ -p'^2 - \gamma(p' - p)^2 - \beta a p'^2 \right\}.$$

The optimality condition implies

$$p_t = \frac{\gamma}{1 + \gamma + \beta a} p_{t-1}$$

and

$$V(p_{t-1}) = -\frac{\gamma(1 + \beta a)}{1 + \gamma + \beta a} p_{t-1}^2.$$

Thus, parameter  $a$  indeed solves the equation

$$a = \frac{\gamma(1 + \beta a)}{1 + \gamma + \beta a}.$$

### Problem 3

1. The Bellman equation is

$$V(K) = \max_{K'} \left\{ AK^\alpha - (K' - (1 - \delta)K) - \frac{\phi(K' - (1 - \delta)K)^2}{2K} + \beta V(K') \right\}.$$

2. The first-order condition

$$-1 - \phi \frac{I}{K} + \beta V'(K') = 0$$

and the envelope condition

$$V'(K) = \alpha A K^{\alpha-1} + (1 - \delta) + \phi(1 - \delta) \frac{I}{K} + \frac{\phi}{2} \frac{I^2}{K^2}$$

can be combined to obtain

$$1 + \phi \frac{I_t}{K_t} = \beta \left[ \alpha A K_{t+1}^{\alpha-1} + \frac{\phi}{2} \frac{I_{t+1}^2}{K_{t+1}^2} + (1 - \delta) \left( 1 + \phi \frac{I_{t+1}}{K_{t+1}} \right) \right].$$

3. Note that investment and adjustment costs are flows and hence,

$$V(K_t) = \max_{I_t} \left\{ A K_t^\alpha \Delta - I_t \Delta - \frac{\phi}{2} \frac{I_t^2}{K_t} \Delta + e^{-\rho \Delta} V(K_{t+\Delta}) \right\},$$

where the maximization is subject to the law of motion for capital

$$K_{t+\Delta} = (1 - \delta \Delta) K_t + I_t \Delta.$$

4. Approximate the discount factor and rearrange terms

$$\max_{I_t} \left\{ A K_t^\alpha \Delta - I_t \Delta - \frac{\phi}{2} \frac{I_t^2}{K_t} \Delta - \rho V(K_{t+\Delta}) \Delta + V(K_{t+\Delta}) - V(K_t) \right\} = 0.$$

Divide by  $\Delta$  and take the limit  $\Delta \rightarrow 0$ :

$$\rho V(K_t) = \max_{I_t} \left\{ A K_t^\alpha - I_t - \frac{\phi}{2} \frac{I_t^2}{K_t} + V'(K_t) \dot{K}_t \right\}.$$

Substitute the capital law of motion and drop time subscripts to obtain the HJB equation:

$$\rho V(K) = \max_I \left\{ A K^\alpha - I - \frac{\phi}{2} \frac{I^2}{K} + V'(K)(I - \delta K) \right\}.$$